Tolerance-aware auditing and a rational interval certificate for a variable-radius circle packing with 26 circles

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Reproducibility repository

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Abstract

We study the problem of placing 26 circles of independently variable radii in the unit square while maximizing the sum of their radii. Public numerical results for this problem are often reported under different feasibility tolerances, so their objective values are not directly comparable. We provide three separate finite-decimal certificates for the contracts $\tau = 10^{-6}$, 10^{-10}, and 0, together with an exact-rational verifier and a hash-authenticated acquisition of public witnesses. Each author certificate ranks first among the complete acquired witnesses valid under its matching rational contract. More importantly, we give a computer-assisted proof for the real 78-contact configuration near the strict finite-decimal witness. Exact rational interval arithmetic proves a strict Krawczyk inclusion for the 78 active polynomial gap equations, strict feasibility of all 351 inactive geometric constraints, and a second Krawczyk inclusion for the Karush–Kuhn–Tucker multipliers. All 78 multipliers are positive. The active gaps are therefore local coordinates, and the configuration is a strict local maximizer. The argument uses only the Python standard library on its verification path. This is a local theorem and a reproducibility result, not a proof of global optimality or a new packing record.

1 Introduction

For centers (x_i, y_i) and radii $r_i > 0$, $0 \le i < 26$, consider

$$\max \; f(x) = \sum_{i=0}^{25} r_i \tag{1}$$

subject to containment in $[0, 1]^2$ and pairwise non-overlap. The problem is an instance of variable-radius circle packing, with 78 continuous variables and 429 geometric inequalities. It is nonconvex and admits many contact topologies.

Computational reports can become misleading when a higher objective value is obtained by consuming a larger feasibility tolerance. A value accepted at 10^{-6} cannot fairly be ranked against a strict feasible witness without recording the two contracts. Our first contribution is therefore methodological: finite-decimal data are interpreted as exact rational numbers, and every public witness is evaluated separately under a named tolerance.

Our main mathematical result concerns the 78-contact real root near the strict finite-decimal witness. We certify a unique regular root in a rational box of radius 10^{-90} and enclose a positive vector of KKT multipliers. This converts numerical evidence of stationarity into a rigorous proof of strict local optimality.

Interval methods have a substantial history in packing. Markót used interval arithmetic to establish structural optimality for congruent-circle packings in the unit square [3], and interval techniques also appear in the broader computational treatment of square packings [5]. We do not claim that interval certification of circle packing is new in itself. The contribution here is the explicit rational, independently executable certificate for this variable-radius, sum-of-radii configuration, combined with tolerance-aware public-corpus auditing.

The complete reproducibility artifact is archived at Zenodo [1]; the repository cited in Section 6 contains the matching certificate and reports. The public numerical table currently attributes the $n = 26$ value to Haowei Lin [4]; consequently, no new geometry or Packomania record is claimed.

Contributions.

1. We formalize three non-interchangeable feasibility contracts and publish a standard-library exact-rational verifier for the associated finite-decimal witnesses.

2. We provide a reproducible, hash-authenticated acquisition and same-tolerance evaluation of complete public witnesses from an explicit corpus.

3. We prove that the unique real solution of a specified 78-contact system in a rational interval box is a strict local maximizer of (1).

2 Problem and numerical contracts

Write

$$x = (x_0, y_0, r_0, \ldots, x_{25}, y_{25}, r_{25}) \in \mathbb{R}^{78}.$$

For each circle define its four wall gaps

$$x_i - r_i, \quad 1 - x_i - r_i, \quad y_i - r_i, \quad 1 - y_i - r_i,$$

and, for $i < j$, the squared pair gap

$$q_{ij}(x) = (x_i - x_j)^2 + (y_i - y_j)^2 - (r_i + r_j)^2. \tag{2}$$

At zero tolerance, feasibility is equivalent to nonnegative wall gaps, $q_{ij} \geq 0$, and positive radii.

For a rational tolerance $\tau \geq 0$, the repository contract permits wall gaps of at least $-\tau$ and requires

$$(x_i - x_j)^2 + (y_i - y_j)^2 \geq (r_i + r_j - \tau)^2 \tag{3}$$

whenever $r_i + r_j - \tau > 0$. All decisions are made over the rational numbers; square roots are used only to print diagnostic margins. This linear-distance allowance is not asserted to reproduce every upstream binary64 decision at a rounding boundary. Source-native binary64 evaluators are reimplemented and reported separately when available.

Table 1 gives the three primary certificates. The two relaxed witnesses fail when rechecked at zero tolerance, while the last row is a strict feasible rational lower bound.

Each primary verification makes 429 geometric decisions and 26 radius positivity decisions. The strict witness was obtained by rounding a high-precision contact root to 90 decimal places and conservatively reducing all radii by approximately 10^{-75}. Thus the serialized CSV is exactly feasible but is not itself the boundary root certified below.

Table 1: Primary finite-decimal certificates. Scores in different rows are not interchangeable because the feasibility contracts differ.

Contract	Exact-rational recomputed score	Interpretation
10^{-6}	2.63599872089287514	relaxed linear-distance contract
10^{-10}	2.63598308647338795	stricter relaxed contract
0	2.63598308491760778318...	strict rational witness

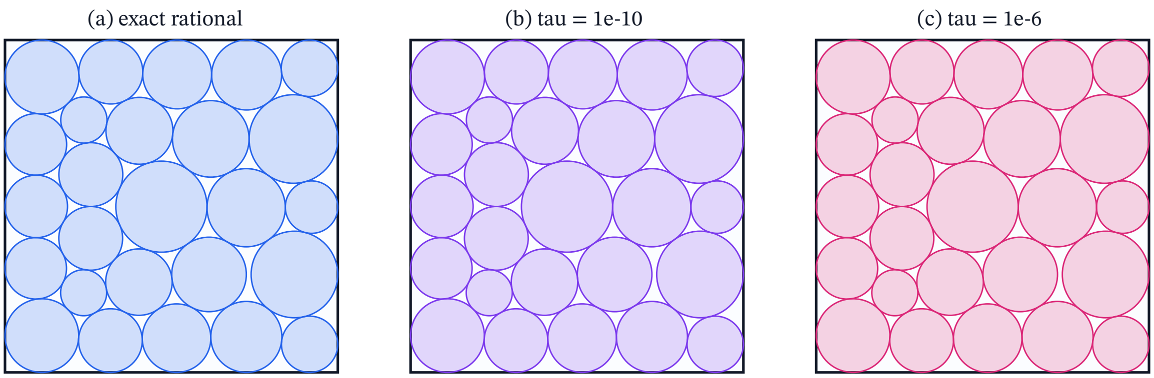

Figure 1: The three serialized witnesses, kept separate by feasibility contract and shown at one common scale. Their geometric differences and tolerance-scale violations are smaller than the plotted line width; a picture therefore cannot replace the contract-aware verifier. The relaxed witnesses consume their stated tolerance and fail at zero tolerance.

3 Reproducible public-corpus audit

The source manifest pins immutable Git artifacts by full commit identifier and SHA-256. Downloaded Python files and notebooks are parsed as data and are never imported or executed. Numeric tokens are retained as decimal strings before conversion to rational numbers. The Packomania text file is not available through an immutable URL; its observed payload is hash-pinned and a changed download makes the default acquisition fail closed.

At the snapshot date, nine upstream artifacts were authenticated and ten complete witnesses were evaluated. Figure 2 reports the five leading valid witnesses under each rational contract; the complete tables, validity matrix, hashes, and exclusions are generated in the archived artifact. Exactly one matching author certificate is included in each panel, and the author's certificates for the other contracts are excluded.

The tolerance labels are audit contracts applied by this work, not necessarily categories declared by the upstream projects. Every complete serialized witness was re-evaluated under all three contracts, and panel membership is determined solely by the resulting exact-rational validity and score. In particular, the Packomania decimal is not treated as an entry originally submitted at $\tau = 10^{-10}$: its serialized value misses strict feasibility by approximately 1.04×10^{-12} and therefore first becomes admissible in our 10^{-10} panel. Conversely, AlphaEvolve v2 passes the reconstructed native binary64 zero-tolerance evaluator but misses our exact-rational zero contract by approximately 5.23×10^{-17}, a rounding-boundary distinction.

Under an independent reimplementation of the EurekAgent native binary64 evaluator with atol $= 10^{-6}$, the matching author witness scores 2.635998720892875 and the public EurekAgent witness scores 2.6359987208598832. This is a source-contract comparison inside the manifested corpus, not a universal rank. AlphaZ-CORAL was authenticated but excluded because its public file computes radii at runtime rather than serializing a complete witness. Numaro and HELIX were recorded as reported claims because no complete downloadable $n = 26$ witness was located during the snapshot.

The three panels make the comparison contract-local. Every row in a panel passes the same exact-rational feasibility test, so its position and score may be compared with the other rows in that panel. Positions must not be compared across panels because the feasible sets differ.

(a) $\tau = 0$ five leading witnesses under one common exact-rational contract

Pos.	Witness	Recomputed score
1	Göther Labs	2.63598308491760778...
2	Jason Liang	2.635983084893
3	Theta 8B-RL Formal	2.63598307738811934
4	ShinkaEvolve	2.63598282664580639
5	Theta AlphaEvolve	2.63586275641369812

(b) $\tau = $ 1e-10 five leading witnesses under one common exact-rational contract

Pos.	Witness	Recomputed score
1	Göther Labs	2.63598308647338795
2	Packomania	2.635983084919
3	AlphaEvolve v2	2.6359830849176068
4	Station	2.63598308491754725
5	Jason Liang	2.635983084893

(c) $\tau = $ 1e-6 five leading witnesses under one common exact-rational contract

Pos.	Witness	Recomputed score
1	Göther Labs	2.63599872089287514
2	Theta 8B-RL	2.63598566124089912
3	Hyra	2.63598309510684482
4	Packomania	2.635983084919
5	AlphaEvolve v2	2.6359830849176068

Göther Labs; compare positions only within the same panel. Other author contracts are excluded.

Figure 2: Tolerance-separated positions in the explicitly manifested corpus. Each panel contains the five leading complete witnesses valid under one common exact-rational contract. The highlighted Göther Labs row is the only author certificate in its panel. Scores are aligned at the decimal point; ellipses truncate displayed digits only. Panel positions are not universal leaderboard claims and are not compared across tolerances. Panel membership reflects our re-evaluation, not an upstream tolerance designation.

Within this deliberately bounded corpus it is accurate to describe each matching author certificate as the *best exact-rationally re-evaluated witness in the manifested corpus*. The unqualified phrase "best-known" would be stronger than the evidence: the corpus is not exhaustive, and reported claims without complete downloadable witnesses cannot be independently ranked.

4 The active polynomial system

The strict witness identifies 78 active constraints: 58 circle–circle contacts and 20 wall contacts. Let

$$g : \mathbb{R}^{78} \longrightarrow \mathbb{R}^{78}$$

collect the corresponding wall gaps and squared pair gaps. Since wall gaps are affine and (2) is quadratic, g is polynomial and its Jacobian $J = Dg$ is affine. This avoids transcendental interval extensions.

The proof certificate stores a rational midpoint m, a rational box $X = m + [-\rho, \rho]^{78}$ with $\rho = 10^{-90}$, and rational preconditioners. For a preconditioner C, the interval Krawczyk image is

$$K(X) = m - Cg(m) + (I - CJ(X))(X - m). \tag{4}$$

All interval endpoints and all acceptance comparisons are computed using arbitrary-size integers and rational fractions. Floating-point libraries are used only by an optional certificate-generation mode; they do not participate in verification.

The exact computation gives

$$K(X) \subset \operatorname{int}(X), \tag{5}$$

$$\max_k \frac{\sup |K_k(X) - m_k|}{\rho} < 8.551052960068879 \times 10^{-15}, \tag{6}$$

$$\|I - CJ(X)\|_\infty < 8.551052960068879 \times 10^{-15}. \tag{7}$$

The Krawczyk theorem [2] therefore gives a unique zero $x^* \in X$. The norm bound also proves that every Jacobian represented in the box is nonsingular. Direct rational interval evaluation proves that all 351 inactive geometric gaps are strict and all radii are positive:

$$\min(\text{inactive polynomial gaps}) > 0.0071877548062774697881043072924253371679, \tag{8}$$

$$\min_i r_i > 0.0691806763572344820974407544002623682677. \tag{9}$$

5 Dual certificate and strict local optimality

Use the convention that feasible gaps satisfy $g \geq 0$. Stationarity for the maximization problem is

$$\nabla f(x^*) + J(x^*)^T \lambda = 0. \tag{10}$$

A second rational Krawczyk test encloses the solution of this interval linear system in a dual box of radius 10^{-10}. Its maximum component inclusion ratio is below $7.677651441669678 \times 10^{-6}$, and

$$\min_k \lambda_k > 0.020825602106288. \tag{11}$$

Theorem 1 (Certified strict local maximum)**.** *The unique zero x^* of the named 78-contact system in X is a strictly feasible packing with respect to every inactive constraint and is a strict local maximizer of $f = \sum_i r_i$ over all feasible packings of 26 variable-radius circles in the unit square.*

Proof. Equations (5)–(7) prove existence, uniqueness, and nonsingularity of $J(x^*)$. Equations (8) and (9) show that the enclosed root is a packing and that no inactive constraint becomes active at the root.

By the inverse function theorem, $z = g(x)$ is a system of local coordinates in a neighborhood of x^*. Restrict the coordinate image, if necessary, to a convex ball about 0, and set $F(z) = f(g^{-1}(z))$. From (10),

$$\nabla_z F(0) = J(x^*)^{-T} \nabla f(x^*) = -\lambda.$$

Every component is strictly negative by (11). By continuity, after shrinking the coordinate neighborhood if necessary, every component of $\nabla_z F$ remains negative. Any nearby feasible packing has $z \geq 0$; if it differs from x^*, then $z \neq 0$. Integrating along the segment from 0 to z yields

$$F(z) - F(0) = \int_0^1 \nabla F(tz) \cdot z \, dt < 0.$$

The inactive constraints remain strict in a sufficiently small neighborhood, so this covers all nearby feasible packings. Hence x^* is a strict local maximizer. $\square$

No Hessian test is required: the active gradients form a basis and the critical cone is trivial. Approximate numerical rank and multiplier signs alone would not justify the theorem; the primal and dual interval inclusions are essential.

The certified score interval at the contact root implies the more compact outward-rounded enclosure

$$f(x^*) > 2.635983084917607783186569485443481730396676798274 4,$$
$$f(x^*) < 2.635983084917607783186569485443481730396676798274 5.$$

The substantially longer rational endpoints are retained in the machine-readable report. The root score is slightly above the deliberately shrunken finite-decimal witness.

6 Reproducibility

The principal proof can be rerun from the repository root with

```
python3 -S scripts/prove_local_optimum.py
```

where `-S` disables third-party site packages. The command parses all certificate decimals as rational numbers, verifies the primal and dual inclusions, and regenerates the JSON proof report. The full audit, tests, table generation, and visualization run with

```
./verify_all.sh
```

The public corpus can be refreshed online or replayed from an authenticated cache. Third-party programs are never executed. The artifact records the reference software environment, hashes, upstream commits, source licenses, and the provenance boundary of private exploratory material.

The contact graph in Figure 3 is generated from the strict certificate. It contains 58 pair contacts and 20 wall contacts.

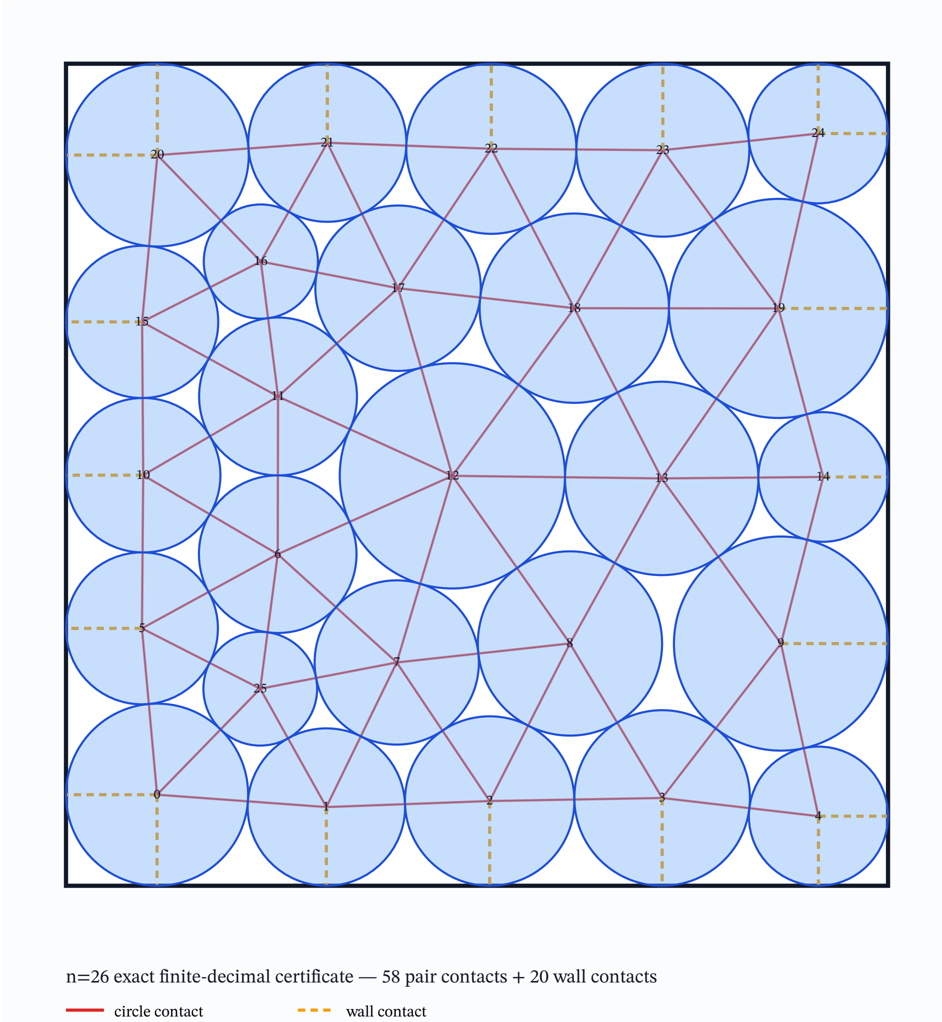

Figure 3: The 26-circle strict witness and its reconstructed contact graph. The interval theorem concerns the nearby exact real contact root.

7 Scope and limitations

Theorem 1 is deliberately local. It neither supplies a global upper bound nor rules out a superior packing with a different contact graph. The configuration was already public, and its score agrees with the current public $n = 26$ value to the displayed precision; we claim neither a new record nor discovery of a new layout. The public-corpus positions cover only the explicit, hash-manifested sources and exclude claims lacking complete downloadable witnesses. Scores belonging to different tolerance contracts must not be compared as if they shared one feasible set.

The method also certifies only one contact root. Applying the same verifier to several values of n or several contact topologies would be needed to support a broader methodological claim. A separate implementation in another interval package would provide useful defense in depth, although the present proof path is already independent of binary floating-point acceptance decisions.

8 Conclusion

Separating numerical contracts resolves an ambiguity that otherwise rewards larger constraint violations with larger apparent objective values. For the strict $n = 26$ configuration, exact rational interval computation goes further: it certifies a unique regular contact root, strict inactive feasibility, and positive KKT multipliers, proving a strict local maximum. The result is a small but concrete computer-assisted theorem with a fully inspectable certificate. Global optimality remains open.

AI assistance disclosure. This work used a Göther Labs AI-assisted research pipeline. Internally coordinated agents contributed to exploratory computation, artifact recovery, implementation, verification, and drafting. Every public claim is backed by inspectable certificates and reproducible checks. The human author is responsible for every claim, proof, citation, and submission; no AI system is an author.

Data and code availability. Source code and certificates are available at `https://github.com/juan-fernandez-gotherlabs/circle-packing-tolerance-audit`. The version 1.2.1 artifact containing the current interval certificate, preprint, and provider-neutral AI-assistance disclosure is being prepared for a versioned Zenodo archive. The preceding version 1.2.0 artifact remains available as DOI 10.5281/zenodo.21864592, and the earlier version 1.1.0 release remains available as DOI 10.5281/zenodo.21853178.

References

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[2] R. Krawczyk, Newton-Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken, *Computing* 4 (1969), 187–201. doi:10.1007/BF02234767.

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